

# Lesson 10: Mains and electrical safety

Electric Circuits · Year 10 · 50 minutes · AQA 6.2.3

## Why this lesson is here

A metal-cased appliance develops a live-to-case fault. The earth path allows a large fault current so the protection disconnects the supply.

## Learning objective

Explain mains hazards and how earthing and fuses protect users.

## Core teaching point

UK mains: ac, about 230 V, 50 Hz. Live brown; neutral blue; earth green/yellow. Live remains dangerous even if a switch is open.

## Equipment and preparation

Three-core cable/plug model, low-voltage ac/dc traces or diagrams; no mains wiring by students.

## Lesson sequence

| Time  | Phase       | Teacher move and evidence                           |
|-------|-------------|-----------------------------------------------------|
| 0-6   | Retrieval   | Identify ac and dc shapes.                          |
| 6-19  | Model       | Teach 230 V, 50 Hz and cable functions.             |
| 19-31 | Fault story | Students order the events in an earthed metal case. |
| 31-43 | Practice    | Tasks 1-4 with causal language.                     |
| 43-50 | Check       | Five-mark safety explanation.                       |

## Workbook task guidance

Task 1: Draw and label a dc p.d.-time line and an ac p.d.-time wave, including axes.

Expected: dc line stays one side of zero; ac wave alternates positive and negative; label p.d. and time.

Task 2: Colour-code and state the purpose of live, neutral and earth wires.

Expected: Live brown supplies alternating p.d.; neutral blue completes circuit; earth green/yellow is safety path.

Task 3: Explain why touching live can shock someone even if the lamp switch is open.

Expected: The open switch does not isolate the live wire from the supply; body to earth can carry current.

Task 4: Explain the sequence from live-to-metal-case fault to the fuse melting or breaker tripping.

Expected: Fault current goes through low-resistance earth path, melts fuse/trips breaker and disconnects live.

## Worked example

A 50 Hz supply completes 50 cycles each second. A dc supply has a fixed direction; an ac supply repeatedly reverses direction.

## AQA-style checkpoint

[5 marks] A kettle has a metal case connected to earth. Explain why its live wire remains dangerous when the switch is open, and how the earth wire protects a user after a fault.

Mark guidance: Live remains at about 230 V relative to earth (1); a body provides a path to earth so current/shock can occur (1); earth gives low-resistance fault path (1); large current flows (1); fuse melts/breaker trips and disconnects live (1).

### **Anticipated misconception**

Neutral normally carries current. Earth carries current only during a fault.

### **Assessment and next step**

Collect one sentence of reasoning and one complete calculation or graph label. Start the next lesson by reteaching the most common error; students who are secure should tackle a changed numerical context.